

M²S³OM-graph

A hybrid deterministic–LLM pipeline for automated metadata crosswalk extraction and graph-grounded interoperability

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37

RDAMSC crosswalks evaluated

51%

Solved by deterministic parsers

993

SSSOM mapping rows

75.2%

DataCite -> DublinCore coverage

20

Metadata standards in the graph

32/45

Artifacts fetched

7

Domain categories represented

01 · BACKGROUND

Metadata Crosswalks are document-based

Mapping specifications are published as human-readable artifacts. No tool harvests them systematically.

- Research Data Alliance (RDA) provides mappings as documentation: <https://rdamsc.bath.ac.uk/mapping-index>
- Mappings(=Crosswalks) live as human-readable artifacts — HTML tables, XSL stylesheets, PDFs, free text.
- Semantic loss is rarely flagged.
- No existing tool harvests these rules systematically across standards.
- Without a machine-executable layer, every conversion is handled by hand at the data infrastructure provider.

ID	DataCite-Property	Dublin Core
1	identifier	dcterms:identifier
1.a	identifierType	Not present in Dublin Core
2	Creator	dcterms:creator
2.1	creatorName	dcterms:creator
2.1.a	nameType	Not present in Dublin Core
2.2	givenName	Not present in Dublin Core

TABLE · DATACITE ↔ DC

II. Dublin Core to MARC Crosswalk (15 elements in version 1.1 and refinements)
Contributor — An entity responsible for making contributions to the resource.
Type(s):
1010 (Added Entry—Uncontrolled Name/Name)

Qualifiers:
Element refinements may be used from the MARC release list for those specified rules that define DC elements include a valid refinement only used for Contributors, use the term in Title or ID. See [MARC Release Terms and Dublin Core](#).

Note: This is a way to specify whether the Contributor is a person or organization because it is not in the DC. If it can reasonably be determined that the contributor is a person or organization, Gaba 700 105a (Added Entry Name) or 710 105a (Added Entry—Corporate Name) may be used.

Coverage: The spatial or temporal topic of the resource, the spatial applicability of the resource, or the mode which the resource is relevant.
Type(s):
500a (General note)

TEXT · DataCite ↔ MARC

02 · APPROACH

Deterministic first, LLM only when needed

Format-specific parsers carry the load. LLMs are reserved for artifacts that genuinely resist structure.

- Deterministic first.** Format-specific parsers handle structured artifacts without any model call → reproducible, auditable- Parsers were coded with Gen-AI (Blablador).
- LLM as fallback.** Blablador (Helmholtz-AI infrastructure) is invoked only when an artifact resists deterministic parsing.
- SSSOM as target.** Every rule is a row in a TSV as Simple Standard for Sharing Ontological Mappings (SSSOM) — community standard, diff-friendly, version-controlled.



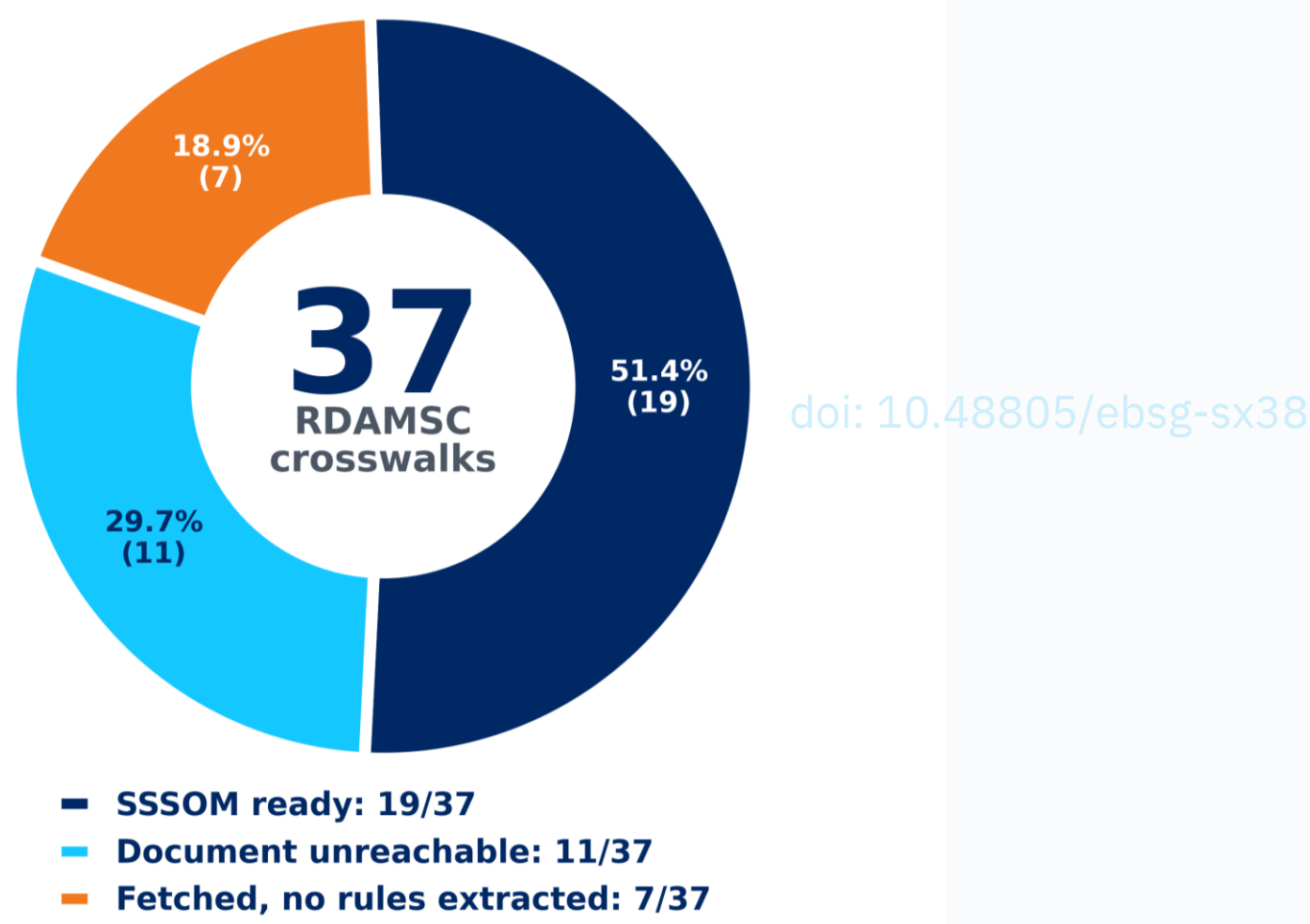
03 · METHOD

Pipeline



04 · RESULTS

Crosswalk outcome distribution

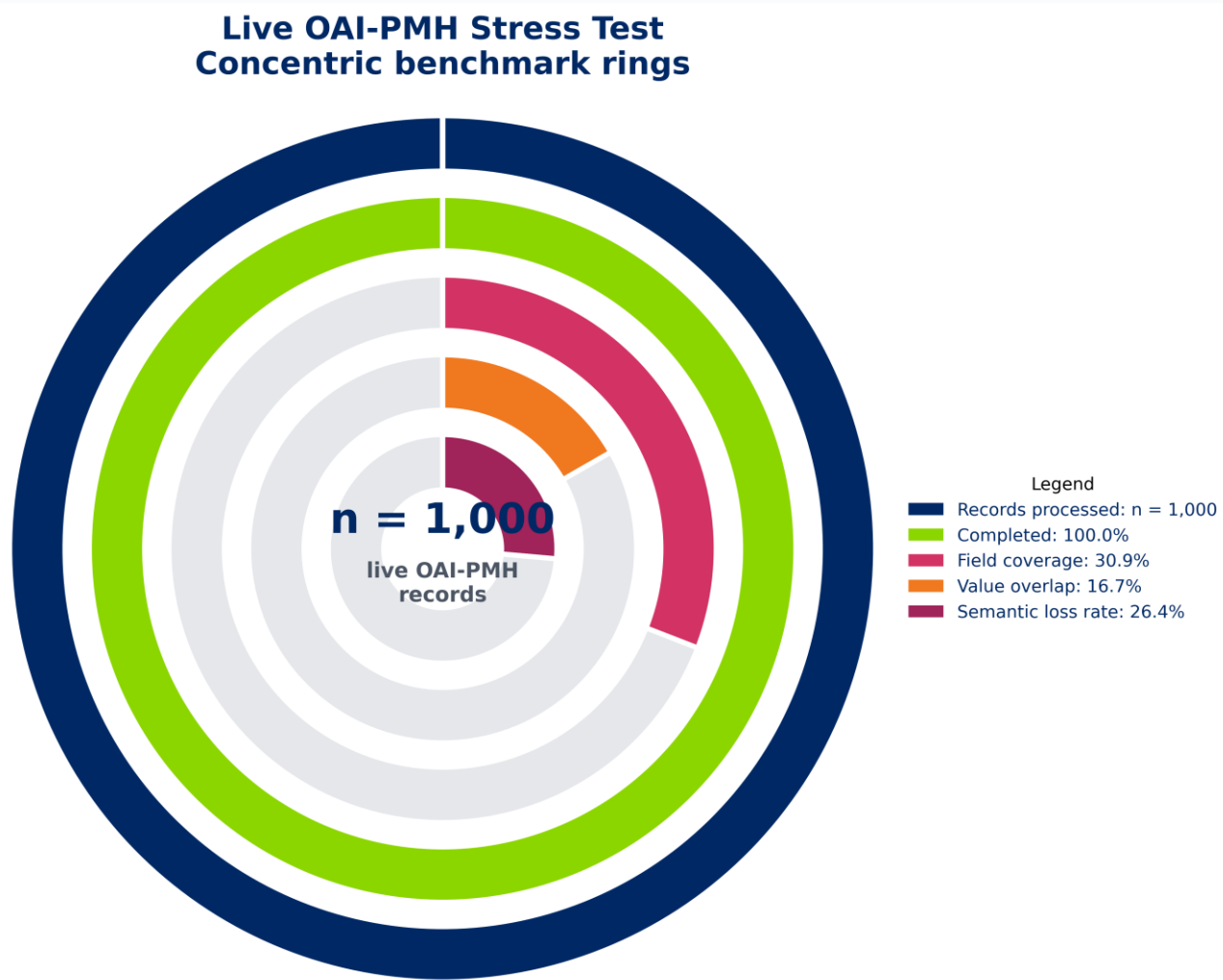


51% of crosswalks have authoritative SSSOM output; 30% blocked by unreachable artifacts.

05 · BENCHMARK

DataCite → Dublin Core

Applying the extracted rules end-to-end on a real n=1000 harvest.

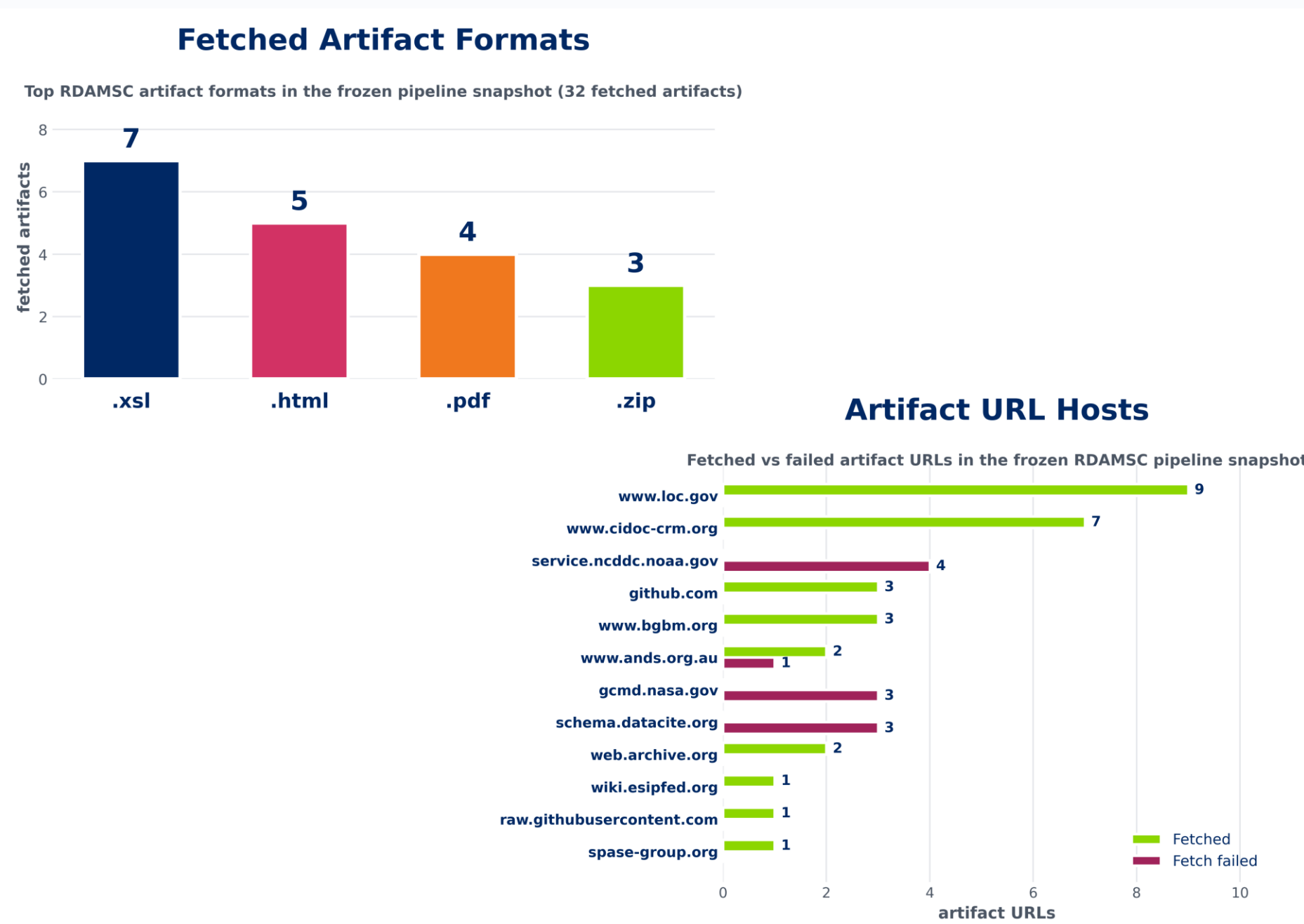


Lower overlap is expected: Dublin Core is a deliberately broad target schema, so rich DataCite metadata is compressed into fewer, less specific fields. The benchmark quantifies this semantic loss rather than indicating pipeline failure.

06 · CORPUS

Artifact formats & hosts

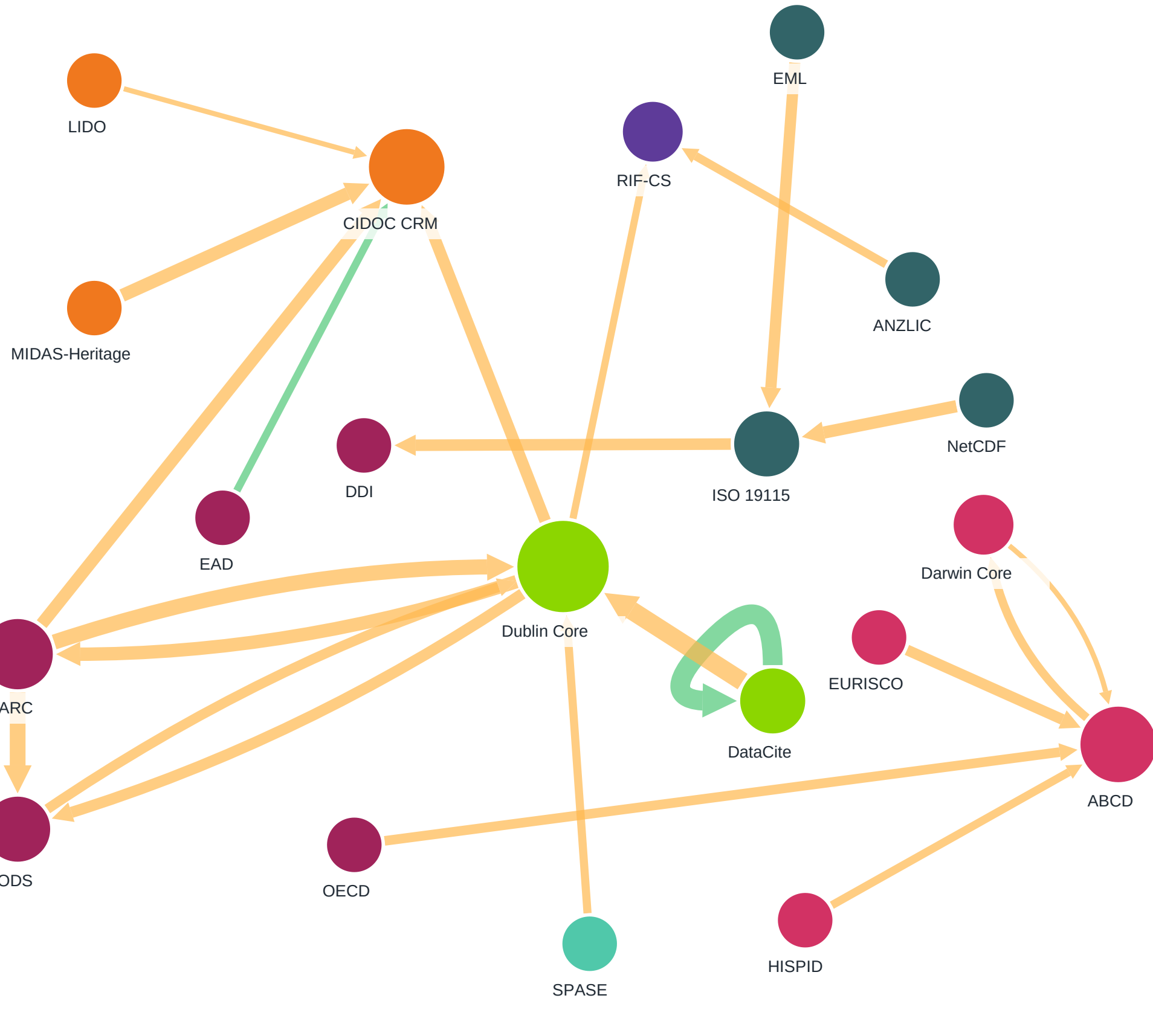
Heterogeneous landscape: XSL and HTML dominate. NASA / NOAA endpoints account for most fetch failures.



07 · GRAPH

Crosswalks as a knowledge graph

Standards are nodes; mapping rules are weighted edges. Direct mappings cover the easy pairs; transitive paths will unlock the long tail.



- Standards become nodes; crosswalks become edges.
- Each edge keeps the mapping rules, confidence, semantic-loss flags, and source evidence.
- The graph makes indirect routes visible, e.g. A → B → C when no direct A → C crosswalk exists.
- Provenance stays attached to the rule, not buried in the original PDF, HTML, or XSL file.

08 · TAKEAWAYS

Conclusions

- Structure beats guesswork: parse what is structured, use LLMs only where documents break down.
- The main bottleneck is not only AI accuracy, but fragile metadata infrastructure: dead links, old formats, and implicit semantics.
- Crosswalks become more useful when they stop being static documents and become executable, versioned SSSOM mappings.
- The graph turns isolated mappings into an interoperability network: standards as nodes, crosswalks as navigable edges.
- Provenance is the difference between “the model said so” and an inspectable mapping rule.
- The benchmark shows the system can run on real records, and makes the remaining semantic gaps measurable.

09 · FUTURE WORK

Graph-RAG over the crosswalk graph

The same evidence chunks that justify extracted rules can propose new ones: retrieval-grounded, not hallucinated.

Transitive paths

Discover A → B → C routes through the graph when no direct crosswalk exists.

Graph-RAG candidates

Retrieve evidence chunks from the graph to propose mappings for uncharted standard pairs.

Confidence from retrieval

Score candidate rules by retrieval quality, not just model self-report.

Curator feedback loop

Accepted / rejected decisions feed back as supervised signal for the extractors.